

Concept 5 | Recurrence and Induction

English Student Edition • Focused formulas and guided practice

Formula opener

Recurrence

$$a_{n+1} = f(a_n)$$

Induction

$$P(1), P(k) \Rightarrow P(k + 1)$$

Divisibility

$$P(k) = mq \Rightarrow P(k + 1) = mq'$$

Classified Practice

Recurrence and Induction

Q001 Written

For the sequence $\{a_n\}$ with $a_1 = 2$ and recurrence $a_{n+1} = 3a_n + (2)$. Find the 2nd through the 4th terms.

Q002 Written

For the sequence $\{a_n\}$ with $a_1 = -6$ and recurrence $a_{n+1} = 2a_n + (3)$. Find the 2nd through the 4th terms.

Classified Practice

Recurrence and Induction

Q003 Written

For the sequence $\{a_n\}$ with $a_1 = -6$ and recurrence $a_{n+1} = 1a_n + (n(n-2))$. Find the 2nd through the 4th terms.

Q004 Written

For the sequence $\{a_n\}$ with $a_1 = -2$ and recurrence $a_{n+1} = 1a_n + (-n)$. Find the 2nd through the 4th terms.

Classified Practice

Recurrence and Induction

Q005 Written

For the sequence $\{a_n\}$ with $a_1 = 2$ and recurrence $a_{n+1} = 1a_n + (2n)$. Find the 2nd through the 4th terms.

Q006 Written

For the sequence $\{a_n\}$ with $a_1 = -2$ and recurrence $a_{n+1} = 1a_n + (3^n)$. Find the 2nd through the 4th terms.

Classified Practice

Recurrence and Induction

Q007 MCQ

Select the correct option: For the sequence $\{a_n\}$ with $a_1 = 3$ and recurrence $a_{n+1} = 3a_n + (0)$. The fourth term is:

Choose the correct option.

A -81

B 81

C 82

D 80

Q008 MCQ

Select the correct option: For the sequence $\{a_n\}$ with $a_1 = 2$ and recurrence $a_{n+1} = 4a_n + (0)$. The fourth term is:

Choose the correct option.

A 133

B 127

C -128

D 128

Classified Practice

Recurrence and Induction

Q009 MCQ

Select the correct option: For the sequence $\{a_n\}$ with $a_1 = 2$ and recurrence $a_{n+1} = 2a_n + (n + 1)$. The fourth term is:

Choose the correct option.

A 34

B 37

C 33

D -34

Q010 MCQ

Select the correct option: For the sequence $\{a_n\}$ with $a_1 = 5$ and recurrence $a_{n+1} = 3a_n + (2n)$. The fourth term is:

Choose the correct option.

A -171

B 171

C 172

D 170

Classified Practice

Recurrence and Induction

Q011 MCQ

Select the correct option: For the sequence $\{a_n\}$ with $a_1 = 1$ and recurrence $a_{n+1} = 1a_n + (4)$. The fourth term is:

Choose the correct option.

A 17

B 11

C -13

D 13

Q012 MCQ

Select the correct option: For the sequence $\{a_n\}$ with $a_1 = 4$ and recurrence $a_{n+1} = -2a_n + (5)$. The fourth term is:

Choose the correct option.

A -17

B -15

C -19

D 17

Classified Practice

Recurrence and Induction

Q013 MCQ

Select the correct option: For the sequence $\{a_n\}$ with $a_1 = 1$ and recurrence $a_{n+1} = -3a_n + (2)$. The fourth term is:

Choose the correct option.

A -15

B 13

C -13

D -12

Q014 MCQ

Select the correct option: For the sequence $\{a_n\}$ with $a_1 = 4$ and recurrence $a_{n+1} = 1a_n + (n^2)$. The fourth term is:

Choose the correct option.

A 22

B 16

C -18

D 18

Classified Practice

Recurrence and Induction

Q015 MCQ

Select the correct option: For the sequence $\{a_n\}$ with $a_1 = 2$ and recurrence $a_{n+1} = 2a_n + (1)$. The fourth term is:

Choose the correct option.

A 20

B -23

C 23

D 28

Q016 MCQ

Select the correct option: For the sequence $\{a_n\}$ with $a_1 = 5$ and recurrence $a_{n+1} = 1a_n + (-3)$. The fourth term is:

Choose the correct option.

A -1

B -7

C 4

D -4

Classified Practice

Recurrence and Induction

Q017 MCQ

Select the correct option: For the sequence $\{a_n\}$ with $a_1 = 3$ and recurrence $a_{n+1} = 1a_n + (n)$. The fourth term is:

Choose the correct option.

A -9

B 9

C 11

D 6

Q018 MCQ

Select the correct option: For the sequence $\{a_n\}$ with $a_1 = 1$ and recurrence $a_{n+1} = 2a_n + (-n)$. The fourth term is:

Choose the correct option.

A -6

B 3

C -3

D 2

Classified Practice

Recurrence and Induction

Q019 Written

For the sequence $\{a_n\}$ with $a_1 = -1$ and recurrence $a_{n+1} = 1a_n + (5)$. Find the general term.

Q020 Written

For the sequence $\{a_n\}$ with $a_1 = 1$ and recurrence $a_{n+1} = 1a_n + (3)$. Find the general term.

Classified Practice

Recurrence and Induction

Q021 Written

For the sequence $\{a_n\}$ with $a_1 = 4$ and recurrence $a_{n+1} = 1a_n + (-2)$. Find the general term.

Q022 Written

For the sequence $\{a_n\}$ with $a_1 = 1$ and recurrence $a_{n+1} = 1a_n + (6)$. Find the general term.

Classified Practice

Recurrence and Induction

Q023 Written

For the sequence $\{a_n\}$ with $a_1 = 3$ and recurrence $a_{n+1} = 1a_n + (-5)$. Find the general term.

Q024 Written

For the sequence $\{a_n\}$ with $a_1 = -2$ and recurrence $a_{n+1} = 1a_n + (3)$. Find the general term.

Classified Practice

Recurrence and Induction

Q025 Written

For the sequence $\{a_n\}$ with $a_1 = 4$ and recurrence $a_{n+1} = 1a_n + (-4)$. Find the general term.

Q026 MCQ

Select the correct option: For the sequence $\{a_n\}$ with $a_1 = 3$ and recurrence $a_{n+1} = 3a_n + (0)$. The general term is:

Choose the correct option.

A $3^n + 2$

B $3^n - 1$

C $3^n + 3$

D $a_n = 3^n$

Classified Practice

Recurrence and Induction

Q027 MCQ

Select the correct option: For the sequence $\{a_n\}$ with $a_1 = 2$ and recurrence $a_{n+1} = 4a_n + (0)$. The general term is:

Choose the correct option.

A $a_n = 2^{2n-1}$

B $\frac{2^{2n+2}}{2}$

C $\frac{2^{2n-2}}{2}$

D $\frac{2^{2n+4}}{2}$

Q028 MCQ

Select the correct option: For the sequence $\{a_n\}$ with $a_1 = 4$ and recurrence $a_{n+1} = 5a_n + (0)$. The general term is:

Choose the correct option.

A $a_n = 4 \cdot 5^{n-1}$

B $\frac{4 \cdot 5^{n+5}}{5}$

C $\frac{4 \cdot 5^n - 5}{5}$

D $\frac{2(2 \cdot 5^n + 5)}{5}$

Classified Practice

Recurrence and Induction

Q029 MCQ

Select the correct option: For the sequence $\{a_n\}$ with $a_1 = 6$ and recurrence $a_{n+1} = 2a_n + (0)$. The general term is:

Choose the correct option.

A $a_n = 3 \cdot 2^n$

B $3 \cdot 2^n + 1$

C $3 \cdot 2^n - 1$

D $3 \cdot 2^n + 2$

Q030 Written

For the sequence $\{a_n\}$ with $a_1 = 2$ and recurrence $a_{n+1} = 3a_n + (0)$. Find the general term.

Classified Practice

Recurrence and Induction

Q031 Written

For the sequence $\{a_n\}$ with $a_1 = 3$ and recurrence $a_{n+1} = 5a_n + (0)$. Find the general term.

Q032 Written

For the sequence $\{a_n\}$ with $a_1 = 4$ and recurrence $a_{n+1} = 3a_n + (0)$. Find the general term.

Classified Practice

Recurrence and Induction

Q033 Written

For the sequence $\{a_n\}$ with $a_1 = 1$ and recurrence $a_{n+1} = 4a_n + (0)$. Find the general term.

Q034 MCQ

Select the correct option: For the sequence $\{a_n\}$ with $a_1 = 1$ and recurrence $a_{n+1} = 1a_n + (4)$. The general term is:

Choose the correct option.

A $4n - 1$

B $a_n = 4n - 3$

C $4n + 1$

D $4n - 5$

Classified Practice

Recurrence and Induction

Q035 MCQ

Select the correct option: For the sequence $\{a_n\}$ with $a_1 = 4$ and recurrence $a_{n+1} = 1a_n + (n^2)$. The general term is:

Choose the correct option.

A $\frac{2n^3 - 3n^2 + n + 36}{6}$

B $a_n = \frac{2n^3 - 3n^2 + n + 24}{6}$

C $\frac{2n^3 - 3n^2 + n + 48}{6}$

D $\frac{2n^3 - 3n^2 + n + 12}{6}$

Q036 MCQ

Select the correct option: For the sequence $\{a_n\}$ with $a_1 = 1$ and recurrence $a_{n+1} = -4a_n + (0)$. The general term is:

Choose the correct option.

A $-\frac{(-4)^n - 8}{4}$

B $-\frac{(-4)^n + 8}{4}$

C $-\frac{(-4)^n - 12}{4}$

D $a_n = (-4)^{n-1}$

Classified Practice

Recurrence and Induction

Q037 MCQ

Select the correct option: For the sequence $\{a_n\}$ with $a_1 = 5$ and recurrence $a_{n+1} = 1a_n + (3^n)$. The general term is:

Choose the correct option.

A $a_n = \frac{3^{n+7}}{2}$

B $\frac{3^{n+9}}{2}$

C $\frac{3^{n^2+3}}{2}$

D $\frac{3^{n^2+11}}{2}$

Q038 Written

For the sequence $\{a_n\}$ with $a_1 = -3$ and recurrence $a_{n+1} = -2a_n + (0)$. Find the general term.

Classified Practice

Recurrence and Induction

Q039 Written

For the sequence $\{a_n\}$ with $a_1 = 2$ and recurrence $a_{n+1} = -5a_n + (0)$. Find the general term.

Q040 Written

For the sequence $\{a_n\}$ with $a_1 = -2$ and recurrence $a_{n+1} = -4a_n + (0)$. Find the general term.

Classified Practice

Recurrence and Induction

Q041 MCQ

Select the correct option: For the sequence $\{a_n\}$ with $a_1 = 5$ and recurrence $a_{n+1} = 1a_n + (-3)$. The general term is:

Choose the correct option.

A $10 - 3n$

B $a_n = 8 - 3n$

C $-3(n - 4)$

D $5 - 3n$

Q042 MCQ

Select the correct option: For the sequence $\{a_n\}$ with $a_1 = 3$ and recurrence $a_{n+1} = 1a_n + (n)$. The general term is:

Choose the correct option.

A $\frac{n(n-1)}{2}$

B $\frac{n^2-n+10}{2}$

C $a_n = \frac{n^2-n+6}{2}$

D $\frac{n^2-n+12}{2}$

Classified Practice

Recurrence and Induction

Q043 MCQ

Select the correct option: For the sequence $\{a_n\}$ with $a_1 = 3$ and recurrence $a_{n+1} = 1a_n + (-5)$. The general term is:

Choose the correct option.

A $-5(n - 2)$

B $a_n = 8 - 5n$

C $12 - 5n$

D $-5(n - 1)$

Q044 MCQ

Select the correct option: For the sequence $\{a_n\}$ with $a_1 = 2$ and recurrence $a_{n+1} = 1a_n + (6)$. The general term is:

Choose the correct option.

A $6n - 7$

B $2(3n - 1)$

C $a_n = 2(3n - 2)$

D $6n - 1$

Classified Practice

Recurrence and Induction

Q045 Written

For the sequence $\{a_n\}$ with $a_1 = 2$ and recurrence $a_{n+1} = 3a_n + (-n)$. Find the 2nd through the 4th terms.

Q046 Written

For the sequence $\{a_n\}$ with $a_1 = 1$ and recurrence $a_{n+1} = -2a_n + (5)$. Find the 2nd through the 4th terms.

Classified Practice

Recurrence and Induction

Q047 Written

For the sequence $\{a_n\}$ with $a_1 = 3$ and recurrence $a_{n+1} = 3a_n + (-n)$. Find the 2nd through the 4th terms.

Classified Practice

Recurrence and Induction

Q048 Challenge

Solve the third-order recurrence $a_n = 6a_{n-1} - 11a_{n-2} + 6a_{n-3}$ for $n \geq 3$ with $a_0 = 0, a_1 = 1, a_2 = 6$.

Classified Practice

Recurrence and Induction

Q049 Written

Find the general term of the sequence $\{a_n\}$ determined by $a_1 = 1$ and $a_{n+1} = a_n + (2n - 3)$.

Q050 Written

Find the general term of the sequence $\{a_n\}$ determined by $a_1 = 1$ and $a_{n+1} = a_n + (3n)$.

Classified Practice

Recurrence and Induction

Q051 Written

Find the general term of the sequence $\{a_n\}$ determined by $a_1 = 2$ and $a_{n+1} = a_n + (2n)$.

Q052 Written

Find the general term of the sequence $\{a_n\}$ determined by $a_1 = 2$ and $a_{n+1} = a_n + (3n^2)$.

Classified Practice

Recurrence and Induction

Q053 Written

Find the general term of the sequence $\{a_n\}$ determined by $a_1 = 2$ and $a_{n+1} = a_n + (3n + 1)$.

Q054 Written

Find the general term of the sequence $\{a_n\}$ determined by $a_1 = 2$ and $a_{n+1} = a_n + (n(2n + 1))$.

Classified Practice

Recurrence and Induction

Q055 MCQ

Select the correct option: If $a_1 = 3$ and $a_{n+1} = a_n + (n^2)$, then a_n is:

Choose the correct option.

A $\frac{(n+1)(2n^2-5n+6)}{6}$

B $\frac{(n+2)(2n^2-7n+15)}{6}$

C $a_n = \frac{2n^3-3n^2+n+18}{6}$

D $\frac{2n^3-3n^2+n+36}{6}$

Q056 MCQ

Select the correct option: If $a_1 = 2$ and $a_{n+1} = a_n + (3n^2)$, then a_n is:

Choose the correct option.

A $a_n = \frac{2n^3 - 3n^2 + n + 4}{2}$

B $\frac{(n+1)(2n^2 - 5n + 6)}{2}$

C $\frac{n(n-1)(2n-1)}{2}$

D $\frac{2n^3 - 3n^2 + n + 8}{2}$

Classified Practice

Recurrence and Induction

Q057 MCQ

Select the correct option: If $a_1 = 2$ and $a_{n+1} = a_n + (n^2 + 1)$, then a_n is:

Choose the correct option.

A $\frac{(n-1)(2n^2-n+6)}{6}$

B $\frac{2n^3-3n^2+7n+18}{6}$

C $a_n = \frac{2n^3-3n^2+7n+6}{6}$

D $\frac{(2n+3)(n^2-3n+8)}{6}$

Q058 MCQ

Select the correct option: If $a_1 = 5$ and $a_{n+1} = a_n + (n(4n - 1))$, then a_n is:

Choose the correct option.

A $\frac{8n^3 - 15n^2 + 7n + 42}{6}$

B $\frac{8n^3 - 15n^2 + 7n + 18}{6}$

C $\frac{8n^3 - 15n^2 + 7n + 48}{6}$

D $a_n = \frac{(n+1)(8n^2 - 23n + 30)}{6}$

Classified Practice

Recurrence and Induction

Q059 MCQ

Select the correct option: If $a_1 = 2$ and $a_{n+1} = a_n + (8n)$, then a_n is:

Choose the correct option.

A $4n(n-1)$

B $4(n^2 - n + 1)$

C $a_n = 2(2n^2 - 2n + 1)$

D $4n^2 - 4n + 5$

Q060 MCQ**Select the correct option: find a_4 when $a_1 = 3$ and $a_{n+1} = a_n + (n^2)$:****Choose the correct option.**

A 17

B 19

C 15

D -17

Classified Practice

Recurrence and Induction

Q061 MCQ

Select the correct option: find a_5 when $a_1 = 2$ and $a_{n+1} = a_n + (3n^2)$:

Choose the correct option.

A 96

B 90

C -92

D 92

Q062 MCQ**Select the correct option: find a_3 when $a_1 = 2$ and $a_{n+1} = a_n + (n^2 + 1)$:****Choose the correct option.**

A 9

B 11

C 7

D -9

Classified Practice

Recurrence and Induction

Q063 MCQ

Select the correct option: find a_6 when $a_1 = 5$ and $a_{n+1} = a_n + (n(4n - 1))$:

Choose the correct option.

A -210

B 210

C 215

D 208

Q064 MCQ

Select the correct option: find a_2 when $a_1 = 2$ and $a_{n+1} = a_n + (8n)$:

Choose the correct option.

A 10

B 12

C 8

D -10

Classified Practice

Recurrence and Induction

Q065 Written

Find the general term of the sequence $\{a_n\}$ determined by $a_1 = 2$ and $a_{n+1} = a_n + (n^2)$.

Q066 MCQ

Select the correct option: If $a_1 = 1$ and $a_{n+1} = a_n + (2n + 1)$, then a_n is:

Choose the correct option.

A $a_n = n^2$

B $n^2 + 1$

C $n^2 - 3$

D $n^2 + 2$

Classified Practice

Recurrence and Induction

Q067 Written

Find the general term of the sequence $\{a_n\}$ determined by $a_1 = 1$ and $a_{n+1} = a_n + (2^n)$.

Q068 Written

Find the general term of the sequence $\{a_n\}$ determined by $a_1 = 2$ and $a_{n+1} = a_n + (-3^n)$.

Classified Practice

Recurrence and Induction

Q069 Written

Find the general term of the sequence $\{a_n\}$ determined by $a_1 = 4$ and $a_{n+1} = a_n + (-4^n)$.

Classified Practice

Recurrence and Induction

Q070 Challenge

A technology startup has a reserve of 400,000 LE, represented by $a_1 = 4$, and burns $0.04n$ hundred-thousand LE each quarter: $a_{n+1} = a_n - 0.04n$. Find an explicit formula for a_n .

Classified Practice

Recurrence and Induction

Q071 Written

Find the general term of $\{a_n\}$ given $a_1 = 1$ and $a_{n+1} = 3a_n + (4)$.

Classified Practice

Recurrence and Induction

Q072 Written

Find the general term of $\{a_n\}$ given $a_1 = 1$ and $a_{n+1} = 5a_n + (8)$.

Classified Practice

Recurrence and Induction

Q073 Written

Find the general term of $\{a_n\}$ given $a_1 = 2$ and $a_{n+1} = 2a_n + (-1)$.

Classified Practice

Recurrence and Induction

Q074 **Written**

Find the general term of $\{a_n\}$ given $a_1 = -3$ and $a_{n+1} = 4a_n + (3)$.

Classified Practice

Recurrence and Induction

Q075 Written

Find the general term of $\{a_n\}$ given $a_1 = 3$ and $a_{n+1} = 3a_n + (-2)$.

Classified Practice

Recurrence and Induction

Q076 Written

Find the general term of $\{a_n\}$ given $a_1 = -2$ and $a_{n+1} = -3a_n + (4)$.

Classified Practice

Recurrence and Induction

Q077 Written

Find the general term of $\{a_n\}$ given $a_1 = 1$ and $a_{n+1} = \frac{1}{3}a_n + (2)$.

Classified Practice

Recurrence and Induction

Q078 Written

Find the general term of $\{a_n\}$ given $a_1 = 2$ and $a_{n+1} = \frac{2}{3}a_n + (-\frac{1}{2})$.

Classified Practice

Recurrence and Induction

Q079 Written

Find the general term of $\{a_n\}$ given $a_1 = 4$ and $a_{n+1} = -2a_n + (3)$.

Classified Practice

Recurrence and Induction

Q080 MCQ

Select the correct option: If $a_1 = 3$ and $a_{n+1} = -2a_n + (5)$, the general term is:

Choose the correct option.

A $-\frac{2(-2)^n - 11}{3}$

B $a_n = -\frac{2(-2)^n - 5}{3}$

C $-\frac{2(-2)^n - 17}{3}$

D $-\frac{2((-2)^n + 2)}{3}$

Classified Practice

Recurrence and Induction

Q081 MCQ

Select the correct option: If $a_1 = 1$ and $a_{n+1} = \frac{2}{3}a_n + (-2)$, the general term is:

Choose the correct option.

A $\frac{3 \cdot 3^{-n}(7 \cdot 2^n - 6 \cdot 3^n)}{2}$

B $\frac{3^{-n}(21 \cdot 2^n - 8 \cdot 3^n)}{2}$

C $a_n = \frac{3 \cdot 3^{-n}(7 \cdot 2^n - 4 \cdot 3^n)}{2}$

D $\frac{3 \cdot 3^{-n}(7 \cdot 2^n - 2 \cdot 3^n)}{2}$

Classified Practice

Recurrence and Induction

Q082 MCQ

Select the correct option: If $a_1 = 2$ and $a_{n+1} = 7a_n + (-4)$, the general term is:

Choose the correct option.

A $\frac{4(7^n+14)}{21}$

B $\frac{4 \cdot 7^n - 49}{21}$

C $\frac{4 \cdot 7^n + 77}{21}$

D $a_n = \frac{2(2 \cdot 7^n + 7)}{21}$

Classified Practice

Recurrence and Induction

Q083 MCQ

Select the correct option: If $a_1 = 6$ and $a_{n+1} = 2a_n + (-5)$, the general term is:

Choose the correct option.

A $\frac{2^n+14}{2}$

B $a_n = \frac{2^n+10}{2}$

C $\frac{2^n+18}{2}$

D $\frac{2^n+4}{2}$

Classified Practice

Recurrence and Induction

Q084 MCQ

Select the correct option: If $a_1 = 5$ and $a_{n+1} = -2a_n + (7)$, the general term is:

Choose the correct option.

A $-\frac{2(2(-2)^n+1)}{3}$

B $-\frac{4(-2)^n-13}{3}$

C $a_n = -\frac{4(-2)^n-7}{3}$

D $-\frac{4((-2)^n-4)}{3}$

Classified Practice

Recurrence and Induction

Q085 MCQ

Select the correct option: If $a_1 = 4$ and $a_{n+1} = -5a_n + (-2)$, the general term is:

Choose the correct option.

A $-\frac{13(-5)^n - 25}{15}$

B $-\frac{13(-5)^n + 50}{15}$

C $-\frac{13(-5)^n - 40}{15}$

D $a_n = -\frac{13(-5)^n + 5}{15}$

Classified Practice

Recurrence and Induction

Q086 MCQ

Select the correct option: If $a_1 = 2$ and $a_{n+1} = -3a_n + (5)$, the general term is:

Choose the correct option.

A $-\frac{(-3)^n - 13}{4}$

B $a_n = -\frac{(-3)^n - 5}{4}$

C $-\frac{(-3)^n - 21}{4}$

D $-\frac{(-3)^n + 7}{4}$

Classified Practice

Recurrence and Induction

Q087 MCQ

Select the correct option: If $a_1 = 1$ and $a_{n+1} = \frac{1}{2}a_n + (7)$, the general term is:

Choose the correct option.

A $2^{-n} (11 \cdot 2^n - 26)$

B $2 \cdot 2^{-n} (8 \cdot 2^n - 13)$

C $a_n = 2 \cdot 2^{-n} (7 \cdot 2^n - 13)$

D $2^{-n} (17 \cdot 2^n - 26)$

Classified Practice

Recurrence and Induction

Q088 MCQ

Select the correct option: If $a_1 = 1$ and $a_{n+1} = 2a_n + (-1)$, the general term is:

Choose the correct option.

A 3

B 0

C 4

D $a_n = 1$

Classified Practice

Recurrence and Induction

Q089 MCQ

Select the correct option: If $a_1 = 3$ and $a_{n+1} = -2a_n + (5)$, the general term is:

Choose the correct option.

A $-\frac{2(-2)^n - 11}{3}$

B $-\frac{2((-2)^n + 2)}{3}$

C $-\frac{2((-2)^n - 7)}{3}$

D $a_n = -\frac{2(-2)^n - 5}{3}$

Classified Practice

Recurrence and Induction

Q090 Challenge

A sequence satisfies $a_{n+1} = 4a_n - 9$ with $a_1 = 2$. Find the equilibrium value, the explicit formula, and a_5 .

Classified Practice

Recurrence and Induction

Q091 Written

Use mathematical induction to prove $\sum_{i=1}^n i = \frac{n(n+1)}{2}$.

Classified Practice

Recurrence and Induction

Q092 Written

Use mathematical induction to prove $\sum_{i=1}^n (2i - 1)^2 = \frac{n(2n-1)(2n+1)}{3}$.

Classified Practice

Recurrence and Induction

Q093 Written

Use mathematical induction to prove $\sum_{i=1}^n i(i+1) = \frac{n(n+1)(n+2)}{3}$.

Classified Practice

Recurrence and Induction

Q094 MCQ

Select the correct option: the right-hand side equivalent to $\sum_{i=1}^n i(i+1)$ is:

Choose the correct option.

A $\frac{n(n+1)(n+2)}{3}$

B $\frac{n^3+3n^2+2n+3}{3}$

C $\frac{n^3+3n^2+2n-3}{3}$

D $\frac{(n+3)(n^2+2)}{3}$

Classified Practice

Recurrence and Induction

Q095 MCQ

Select the correct option: the right-hand side equivalent to $\sum_{i=1}^n 3 \cdot 4^{i-1}$ is:

Choose the correct option.

A $4^n + 1$

B $4^n - 1$

C $4^n + 3$

D $4^n - 2$

Classified Practice

Recurrence and Induction

Q096 MCQ

Select the correct option: the right-hand side equivalent to $\sum_{i=1}^n 2i + 3$ is:

Choose the correct option.

A $n^2 + 4n - 1$

B $n^2 + 4n + 2$

C $n(n + 4)$

D $(n + 1)(n + 3)$

Classified Practice

Recurrence and Induction

Q097 MCQ

Select the correct option: the right-hand side equivalent to $\sum_{i=1}^n 2i^2 + 3i$ is:

Choose the correct option.

A $\frac{4n^3+15n^2+11n+12}{6}$

B $\frac{n(n+1)(4n+11)}{6}$

C $\frac{4n^3+15n^2+11n+24}{6}$

D $\frac{(n+2)(4n^2+7n-3)}{6}$

Classified Practice

Recurrence and Induction

Q098 MCQ

Select the correct option: the right-hand side equivalent to $\sum_{i=1}^n 3i^2 + 2i$ is:

Choose the correct option.

A $\frac{2n^3 + 5n^2 + 3n - 2}{2}$

B $\frac{2n^3 + 5n^2 + 3n + 4}{2}$

C $\frac{n(n+1)(2n+3)}{2}$

D $\frac{2n^3 + 5n^2 + 3n + 6}{2}$

Classified Practice

Recurrence and Induction

Q099 MCQ

Select the correct option: the right-hand side equivalent to $\sum_{i=1}^n 7 \cdot 2^{i-1}$ is:

Choose the correct option.

A $7(2^n - 1)$

B $72^n - 6$

C $72^n - 8$

D $72^n - 5$

Classified Practice

Recurrence and Induction

Q100 MCQ

Select the correct option: the right-hand side equivalent to $\sum_{i=1}^n i(i+1)$ is:

Choose the correct option.

A $\frac{n^3+3n^2+2n-3}{3}$

B $\frac{(n+3)(n^2+2)}{3}$

C $\frac{n(n+1)(n+2)}{3}$

D $\frac{n^3+3n^2+2n+9}{3}$

Classified Practice

Recurrence and Induction

Q101 MCQ

Select the correct option: the right-hand side equivalent to $\sum_{i=1}^n 3 \cdot 4^{i-1}$ is:

Choose the correct option.

A $4^n - 1$

B 4^n

C $4^n - 2$

D $4^n + 1$

Classified Practice

Recurrence and Induction

Q102 MCQ

Select the correct option: the right-hand side equivalent to $\sum_{i=1}^n 2i + 3$ is:

Choose the correct option.

A $n^2 + 4n + 2$

B $n(n + 4)$

C $(n + 2)^2$

D $n^2 + 4n - 1$

Classified Practice

Recurrence and Induction

Q103 Written

Use mathematical induction to prove $\sum_{i=1}^n (2i - 1) = n^2$.

Classified Practice

Recurrence and Induction

Q104 **Written**

Use mathematical induction to prove $\sum_{i=1}^n i(i+1) = \frac{n(n+1)(n+2)}{3}$.

Classified Practice

Recurrence and Induction

Q105 Written

Use mathematical induction to prove $\sum_{i=1}^n i(i+2) = \frac{n(n+1)(2n+7)}{6}$.

Classified Practice

Recurrence and Induction

Q106 Written

Use mathematical induction to prove $\sum_{i=1}^n 3 \cdot 4^{i-1} = 4^n - 1$.

Classified Practice

Recurrence and Induction

Q107 Written

Use mathematical induction to prove $\sum_{i=0}^n 10^i = \frac{10^{n+1}-10}{9}$.

Classified Practice

Recurrence and Induction

Q108 Challenge

Use mathematical induction to prove $\sum_{i=1}^n (i^2 + i + 1) = \frac{n(n^2+3n+5)}{3}$.

Classified Practice

Recurrence and Induction

Q109 **Written****Prove by mathematical induction that $2^n > 3n$ for $n \geq 4$.**

Classified Practice

Recurrence and Induction

Q110 Written

Prove by mathematical induction that $3^n > 5n + 1$ for $n \geq 3$.

Classified Practice

Recurrence and Induction

Q111 **Written****Prove by mathematical induction that $3^n > 2n$ for $n \geq 1$.**

Classified Practice

Recurrence and Induction

Q112 Written

Prove by mathematical induction that $2^n > 1n + 2$ for $n \geq 3$.

Classified Practice

Recurrence and Induction

Q113 Written

Prove by mathematical induction that $3^n > 5n + 1$ for $n \geq 4$.

Classified Practice

Recurrence and Induction

Q114 **Written**

Prove by mathematical induction that $3^n > 5n + 2$ for $n \geq 4$.

Classified Practice

Recurrence and Induction

Q115 Written

Prove by mathematical induction that $4^n > 2n + 3$ for $n \geq 4$.

Classified Practice

Recurrence and Induction

Q116 MCQ

Select the correct option: which inequality is proved for $n \geq 3$?:**Choose the correct option.**

A $2^n \geq 1n + 2$

B $2^n > 2n + 2$

C $2^n + 1 > 1n + 2$

D $2^n > 1n + 2$

Classified Practice

Recurrence and Induction

Q117 MCQ

Select the correct option: which inequality is proved for $n \geq 2$:**Choose the correct option.**

A $3^n > 2n$

B $3^n \geq 2n + 0$

C $3^n > 3n + 0$

D $3^n + 1 > 2n + 0$

Classified Practice

Recurrence and Induction

Q118 MCQ

Select the correct option: which inequality is proved for $n \geq 4$?:**Choose the correct option.**

A $3^n + 1 > 5n + 1$

B $3^n > 5n + 1$

C $3^n \geq 5n + 1$

D $3^n > 6n + 1$

Classified Practice

Recurrence and Induction

Q119 MCQ

Select the correct option: which inequality is proved for $n \geq 3$?:**Choose the correct option.**

A $3^n \geq 3n + 0$

B $3^n > 4n + 0$

C $3^n + 1 > 3n + 0$

D $3^n > 3n$

Classified Practice

Recurrence and Induction

Q120 MCQ

Select the correct option: which inequality is proved for $n \geq 4$?:**Choose the correct option.**

A $3^n > 5n + 2$

B $3^n \geq 5n + 2$

C $3^n > 6n + 2$

D $3^n + 1 > 5n + 2$

Classified Practice

Recurrence and Induction

Q121 MCQ

Select the correct option: which inequality is proved for $n \geq 3$?:**Choose the correct option.**

A $4^n + 1 > 3n + 1$

B $4^n > 3n + 1$

C $4^n \geq 3n + 1$

D $4^n > 4n + 1$

Classified Practice

Recurrence and Induction

Q122 MCQ

Select the correct option: which inequality is proved for $n \geq 4$?:**Choose the correct option.**

A $4^n \geq 2n + 3$

B $4^n > 3n + 3$

C $4^n + 1 > 2n + 3$

D $4^n > 2n + 3$

Classified Practice

Recurrence and Induction

Q123 MCQ

Select the correct option: which inequality is proved for $n \geq 4$:**Choose the correct option.**

A $4^n > 4n + 1$

B $4^n \geq 4n + 1$

C $4^n > 5n + 1$

D $4^n + 1 > 4n + 1$

Classified Practice

Recurrence and Induction

Q124 **Written**

Prove by mathematical induction that $2^n > 3n + 1$ for $n \geq 4$.

Classified Practice

Recurrence and Induction

Q125 MCQ

Select the correct option: which inequality is proved for $n \geq 3$?:**Choose the correct option.**

A $4^n + 1 > 3n + 0$

B $4^n > 3n$

C $4^n \geq 3n + 0$

D $4^n > 4n + 0$

Classified Practice

Recurrence and Induction

Q126 **Challenge****When $h > 0$ and $n \geq 3$, prove $(1 + h)^n > 1 + nh^2$.**

Classified Practice

Recurrence and Induction

Q127 Written

Prove that $n^3 + (n + 1)^3 + (n + 2)^3$ is a multiple of 9 for every natural number n , using mathematical induction.

Classified Practice

Recurrence and Induction

Q128 Written

Prove that $n^3 + 2n$ is a multiple of 3 for every natural number n , using mathematical induction.

Classified Practice

Recurrence and Induction

Q129 **Written****Prove by mathematical induction that $n^2 + n$ is a multiple of 2.**

Classified Practice

Recurrence and Induction

Q130 Written

Prove by mathematical induction that $n^2 + 5n + 8$ is a multiple of 2.

Classified Practice

Recurrence and Induction

Q131 **Written**

Prove by mathematical induction that $n^2 + 7n + 12$ is a multiple of 2.

Classified Practice

Recurrence and Induction

Q132 Written

Prove by mathematical induction that $n^2 + 11n + 32$ is a multiple of 2.

Classified Practice

Recurrence and Induction

Q133 MCQ

Select the correct option: which positive integer is guaranteed to divide $n^2 + n$ for every natural number n ?:

Choose the correct option.

A 2

B 3

C 4

D 5

Classified Practice

Recurrence and Induction

Q134 MCQ

Select the correct option: which positive integer is guaranteed to divide $n^2 + 5n + 8$ for every natural number n ?:

Choose the correct option.

A 3

B 4

C 5

D 2

Classified Practice

Recurrence and Induction

Q135 MCQ

Select the correct option: which positive integer is guaranteed to divide $n^2 + 7n + 12$ for every natural number n ?:

Choose the correct option.

A 4

B 5

C 2

D 3

Classified Practice

Recurrence and Induction

Q136 MCQ

Select the correct option: which positive integer is guaranteed to divide $n^2 + 11n + 32$ for every natural number n ?:

Choose the correct option.

A 4

B 5

C 2

D 3

Classified Practice

Recurrence and Induction

Q137 MCQ

Select the correct option: which positive integer is guaranteed to divide $n^3 + 3n^2 + 2n$ for every natural number n ?:

Choose the correct option.

A 9

B 6

C 7

D 8

Classified Practice

Recurrence and Induction

Q138 MCQ

Select the correct option: which positive integer is guaranteed to divide $n^3 + 9n^2 + 26n + 30$ for every natural number n ?:

Choose the correct option.

A 6

B 7

C 8

D 9

Classified Practice

Recurrence and Induction

Q139 MCQ

Select the correct option: which positive integer is guaranteed to divide $n^3 + 12n^2 + 47n + 60$ for every natural number n ?:

Choose the correct option.

A 7

B 8

C 9

D 6

Classified Practice

Recurrence and Induction

Q140 MCQ

Select the correct option: which positive integer is guaranteed to divide $n^3 + 18n^2 + 107n + 216$ for every natural number n ?:

Choose the correct option.

A 7

B 8

C 9

D 6

Classified Practice

Recurrence and Induction

Q141 MCQ

Select the correct option: which positive integer is guaranteed to divide $n^2 + n + 2$ for every natural number n ?:

Choose the correct option.

A 3

B 4

C 5

D 2

Classified Practice

Recurrence and Induction

Q142 **Written**

Prove by mathematical induction that $2n^3 - 3n^2 + n$ is a multiple of 6.

Classified Practice

Recurrence and Induction

Q143 **Written****Prove by mathematical induction that $4n^3 - n$ is a multiple of 3.**

Classified Practice

Recurrence and Induction

Q144 Challenge

For any integer $n \geq 1$, prove that $(n + 1)^n - 1$ is a multiple of n^2 .

Classified Practice

Recurrence and Induction

Q145 Written

For the sequence determined by $a_1 = 3$ and $a_{n+1} = \frac{a_n^2 - 1}{n+1}$, find a_2, a_3, a_4 and estimate the general term.

Q146 Written

For the sequence determined by $a_1 = -1$ and $a_{n+1} = a_n^2 + 2na_n - 2$, find a_2, a_3, a_4 and estimate the general term.

Classified Practice

Recurrence and Induction

Q147 Written

For the sequence determined by $a_1 = 1$ and $a_{n+1} = a_n^2 - n^2 a_n + (n+1)^2$, find a_2, a_3, a_4 and estimate the general term.

Q148 Written

For the sequence determined by $a_1 = 1$ and $a_{n+1} = \frac{a_n - 4}{a_n - 3}$, find a_2, a_3, a_4 and estimate the general term.

Classified Practice

Recurrence and Induction

Q149 Written

For a sequence with $a_1 = 3$ and $a_{n+1} = \frac{a_n^2 - 1}{n+1}$, prove by mathematical induction that $a_n = n + 2$.

Classified Practice

Recurrence and Induction

Q150 MCQ

Select the correct option: for $a_1 = 1$ and $a_{n+1} = a_n + 2$, the general term is:

Choose the correct option.

A $a_n = 2n$

B $a_n = 2(n - 1)$

C $a_n = 1 - 2n$

D $a_n = 2n - 1$

Classified Practice

Recurrence and Induction

Q151 MCQ

Select the correct option: for $a_1 = 2$ and $a_{n+1} = a_n + 2$, the general term is:

Choose the correct option.

A $a_n = 2n - 1$

B $a_n = -2n$

C $a_n = 2n$

D $a_n = 2n + 1$

Classified Practice

Recurrence and Induction

Q152 MCQ

Select the correct option: for $a_1 = 3$ and $a_{n+1} = a_n + 2$, the general term is:

Choose the correct option.

A $a_n = -2n - 1$

B $a_n = 2n + 1$

C $a_n = 2(n + 1)$

D $a_n = 2n$

Classified Practice

Recurrence and Induction

Q153 MCQ

Select the correct option: for $a_1 = 4$ and $a_{n+1} = a_n + 2$, the general term is:

Choose the correct option.

A $a_n = 2(n + 1)$

B $a_n = 2n + 3$

C $a_n = 2n + 1$

D $a_n = -2(n + 1)$

Classified Practice

Recurrence and Induction

Q154 MCQ

Select the correct option: for $a_1 = 1$ and $a_{n+1} = a_n + 3$, the general term is:

Choose the correct option.

A $a_n = 3n - 1$

B $a_n = 3(n - 1)$

C $a_n = 2 - 3n$

D $a_n = 3n - 2$

Classified Practice

Recurrence and Induction

Q155 MCQ

Select the correct option: for $a_1 = 2$ and $a_{n+1} = a_n + 3$, the general term is:

Choose the correct option.

A $a_n = 3n - 2$

B $a_n = 1 - 3n$

C $a_n = 3n - 1$

D $a_n = 3n$

Classified Practice

Recurrence and Induction

Q156 MCQ

Select the correct option: for $a_1 = 3$ and $a_{n+1} = a_n + 3$, the general term is:

Choose the correct option.

A $a_n = -3n$

B $a_n = 3n$

C $a_n = 3n + 1$

D $a_n = 3n - 1$

Classified Practice

Recurrence and Induction

Q157 MCQ

Select the correct option: for $a_1 = 4$ and $a_{n+1} = a_n + 3$, the general term is:

Choose the correct option.

A $a_n = 3n + 1$

B $a_n = 3n + 2$

C $a_n = 3n$

D $a_n = -3n - 1$

Classified Practice

Recurrence and Induction

Q158 **Written**

For a sequence with $a_1 = 1$ and $a_{n+1} = a_n^2 - n^2 a_n + (n+1)^2$, prove by mathematical induction that $a_n = n^2$.

Classified Practice

Recurrence and Induction

Q159 Written

For a sequence with $a_1 = -1$ and $a_{n+1} = a_n^2 + 2na_n - 2$, prove by mathematical induction that $a_n = 1 - 2n$.

Classified Practice

Recurrence and Induction

Q160 MCQ

Select the correct option: for $a_1 = 1$ and $a_{n+1} = 2a_n + 0$, the general term is:

Choose the correct option.

A $a_n = \frac{2^n - 2}{2}$

B $a_n = -2^{n-1}$

C $a_n = 2^{n-1}$

D $a_n = \frac{2^n + 2}{2}$

Classified Practice

Recurrence and Induction

Q161 Written

For a sequence with $a_1 = 1$ and $a_{n+1} = \frac{a_n - 4}{a_n - 3}$, prove by mathematical induction that $a_n = \frac{2n-1}{n}$.

Classified Practice

Recurrence and Induction

Q162 Challenge

Three numbers A, B, C in that order form a geometric sequence with common ratio r . If $A, 2B, C$ in that order form an arithmetic sequence, determine the possible values of r .

Classified Practice

Recurrence and Induction

Quiz MCQ 1 **Concept Quiz · MCQ**

Select the correct option: For the sequence $\{a_n\}$ with $a_1 = 2$ and recurrence $a_{n+1} = 1a_n + (4)$. The general term is:

Choose the correct option.

A $4n$

B $4n - 5$

C $4n + 1$

D $a_n = 2(2n - 1)$

Classified Practice

Recurrence and Induction

Quiz MCQ 2 Concept Quiz · MCQ

Select the correct option: If $a_1 = 2$ and $a_{n+1} = -2a_n + (3)$, the general term is:

Choose the correct option.

A $a_n = -\frac{(-2)^n - 2}{2}$

B $-\frac{(-2)^n - 4}{2}$

C $-\frac{(-2)^n + 4}{2}$

D $-\frac{(-2)^n - 6}{2}$

Classified Practice

Recurrence and Induction

Quiz MCQ 3 **Concept Quiz · MCQ**

Select the correct option: which inequality is proved for $n \geq 4$?

Choose the correct option.

A $2^n + 1 > 3n + 1$

B $2^n > 3n + 1$

C $2^n \geq 3n + 1$

D $2^n > 4n + 1$

Classified Practice

Recurrence and Induction

Quiz MCQ 4 [Concept Quiz · MCQ](#)

Select the correct option: for $a_1 = 4$ and $a_{n+1} = a_n + 2$, the general term is:

Choose the correct option.

A $a_n = 2n + 1$

B $a_n = -2(n + 1)$

C $a_n = 2(n + 1)$

D $a_n = 2n + 3$

Classified Practice

Recurrence and Induction

Quiz WRITTEN 5 Concept Quiz · Written

For the sequence $\{a_n\}$ with $a_1 = 2$ and recurrence $a_{n+1} = 1a_n + (5)$. Find the general term.

Classified Practice

Recurrence and Induction

Quiz WRITTEN 6 Concept Quiz · Written

Find the general term of $\{a_n\}$ given $a_1 = 1$ and $a_{n+1} = 5a_n + (8)$.

Classified Practice

Recurrence and Induction

Quiz WRITTEN 7 Concept Quiz · Written

Prove by mathematical induction that $2^n > 4n + 1$ for $n \geq 4$.

Classified Practice

Recurrence and Induction

Quiz WRITTEN 8 Concept Quiz · Written

For a sequence with $a_1 = 5$ and $a_{n+1} = a_n + 1(2n + 1) + 2$, prove by mathematical induction that $a_n = n^2 + 2n + 2$.

Answer key

Use the question number shown on each page.

Q001 $a_2 = 8, a_3 = 26, a_4 = 80$

Q002 $a_2 = -9, a_3 = -15, a_4 = -27$

Q003 $a_2 = -7, a_3 = -7, a_4 = -4$

Q004 $a_2 = -3, a_3 = -5, a_4 = -8$

Q005 $a_2 = 4, a_3 = 8, a_4 = 14$

Q006 $a_2 = 1, a_3 = 10, a_4 = 37$

Q007 B

Q008 D

Q009 A

Q010 B

Q011 D

Q012 A

Q013 C

Q014 D

Q015 C

Q016 D

Q017 B

Q018 C

Q019 $a_n = 5n - 6$

Q020 $a_n = 3n - 2$

Q021 $a_n = -2(n - 3)$

Q022 $a_n = 6n - 5$

Q023 $a_n = 8 - 5n$

Q024 $a_n = 3n - 5$

Q025 $a_n = -4(n - 2)$

Q026 D

Q027 A

Q028 A

Q029 A

Q030 $a_n = 2 \cdot 3^{n-1}$

Q031 $a_n = 3 \cdot 5^{n-1}$

Q032 $a_n = 4 \cdot 3^{n-1}$

Q033 $a_n = 2^{2n-2}$

Q034 B

Q035 B

Q036 D

Q037 A

Q038 $a_n = \frac{3(-2)^n}{2}$

Q039 $a_n = 2(-5)^{n-1}$

Q040 $a_n = \frac{(-4)^n}{2}$

Q041 B

Q042 C

Q043 B

Q044 C

Q045 $a_2 = 5, a_3 = 13, a_4 = 36$

Q046 $a_2 = 3, a_3 = -1, a_4 = 7$

Q047 $a_2 = 8, a_3 = 22, a_4 = 63$

Q048 $a_n = -\frac{4 \cdot 2^n - 3 \cdot 3^n - 1}{2}$

Q049 $a_n = (n - 2)^2$

Q050 $a_n = \frac{3n^2 - 3n + 2}{2}$

Q051 $a_n = n^2 - n + 2$

Q052 $a_n = \frac{2n^3 - 3n^2 + n + 4}{2}$

Q053 $a_n = \frac{3n^2 - n + 2}{2}$

Q054 $a_n = \frac{4n^3 - 3n^2 - n + 12}{6}$

Q055 C

Q056 A

Q057 C

Q058 D

Q059 C

Q060 A

Q061 D

Q062 A

Q063 B

Q064 A

Q065 $a_n = \frac{2n^3 - 3n^2 + n + 12}{6}$

Q066 A

Q067 $a_n = 2^n - 1$

Q068 $a_n = -\frac{3^n - 7}{2}$

Q069 $a_n = -\frac{4^n - 16}{3}$

Q070 $a_n = -\frac{n^2 - n - 200}{50}$

Q071 $a_n = 3^n - 2$

Q072 $a_n = \frac{3 \cdot 5^n - 10}{5}$

Q073 $a_n = \frac{2^n + 2}{2}$

Q074 $a_n = -\frac{4^n + 2}{2}$

Q075 $a_n = \frac{2 \cdot 3^n + 3}{3}$

Q076 $a_n = (-3)^n + 1$

Q077 $a_n = 3 \cdot 3^{-n} (3^n - 2)$

Q078 $a_n = \frac{3 \cdot 3^{-n} (7 \cdot 2^n - 2 \cdot 3^n)}{4}$

Q079 $a_n = -\frac{3(-2)^n - 2}{2}$

Q080 B

Q081 C

Q082 D

Q083 B

Q084 C

Q085 D

Q086 B

Q087 C

Q088 D

Q089 D

Q090 $c = 3, a_n = -\frac{2^{2n} - 12}{4}, a_5 = -253$

Q091 $\frac{n(n+1)}{2}$

Q092 $\frac{n(2n-1)(2n+1)}{3}$

Q093 $\frac{n(n+1)(n+2)}{3}$

Q094 A

Q095 B	Q113 $3^n > 5n + 1$	Q133 A	Q152 B
Q096 C	Q114 $3^n > 5n + 2$	Q134 D	Q153 A
Q097 B	Q115 $4^n > 2n + 3$	Q135 C	Q154 D
Q098 C	Q116 D	Q136 C	Q155 C
Q099 A	Q117 A	Q137 B	Q156 B
Q100 C	Q118 B	Q138 A	Q157 A
Q101 A	Q119 D	Q139 D	Q158 $a_n = n^2$
Q102 B	Q120 A	Q140 D	Q159 $a_n = 1 - 2n$
Q103 n^2	Q121 B	Q141 D	Q160 C
Q104 $\frac{n(n+1)(n+2)}{3}$	Q122 D	Q142 $6 \mid (2n^3 - 3n^2 + n)$	Q161 $a_n = \frac{2n-1}{n}$
Q105 $\frac{n(n+1)(2n+7)}{6}$	Q123 A	Q143 $3 \mid (4n^3 - n)$	Q162 $r = 2 \pm \sqrt{3}$
Q106 $4^n - 1$	Q124 $2^n > 3n + 1$	Q144 $n^2 \mid [(n+1)^n - 1]$	Quiz MCQ 1 D
Q107 $\frac{10 \cdot 10^n - 1}{9}$	Q125 B	Q145 $a_2 = 4, a_3 = 5, a_4 = 6; a_n = n + 2$	Quiz MCQ 2 A
Q108 $\frac{n(n^2 + 3n + 5)}{3}$	Q126 $(1+h)^n > 1 + nh^2$	Q146 $a_2 = -3, a_3 = -5, a_4 = -7; a_n = 1 - 2n$	Quiz MCQ 3 B
Q109 $2^n > 3n$	Q127 $9 \mid [n^3 + (n+1)^3 + (n+2)^3]$	Q147 $a_2 = 4, a_3 = 9, a_4 = 16; a_n = n^2$	Quiz MCQ 4 C
Q110 $3^n > 5n + 1$	Q128 $3 \mid (n^3 + 2n)$	Q148 $a_2 = \frac{3}{2}, a_3 = \frac{5}{3}, a_4 = \frac{7}{4}; a_n = \frac{2n-1}{n}$	Quiz WRITTEN 5 $a_n = 5n - 3$
Q111 $3^n > 2n$	Q129 $2 \mid (n^2 + n)$	Q149 $a_n = n + 2$	Quiz WRITTEN 6 $a_n = \frac{3 \cdot 5^n - 10}{5}$
Q112 $2^n > 1n + 2$	Q130 $2 \mid (n^2 + 5n + 8)$	Q150 D	Quiz WRITTEN 7 $2^n > 4n + 1$
	Q131 $2 \mid (n^2 + 7n + 12)$	Q151 C	Quiz WRITTEN 8 $a_n = n^2 + 2n + 2$
	Q132 $2 \mid (n^2 + 11n + 32)$		